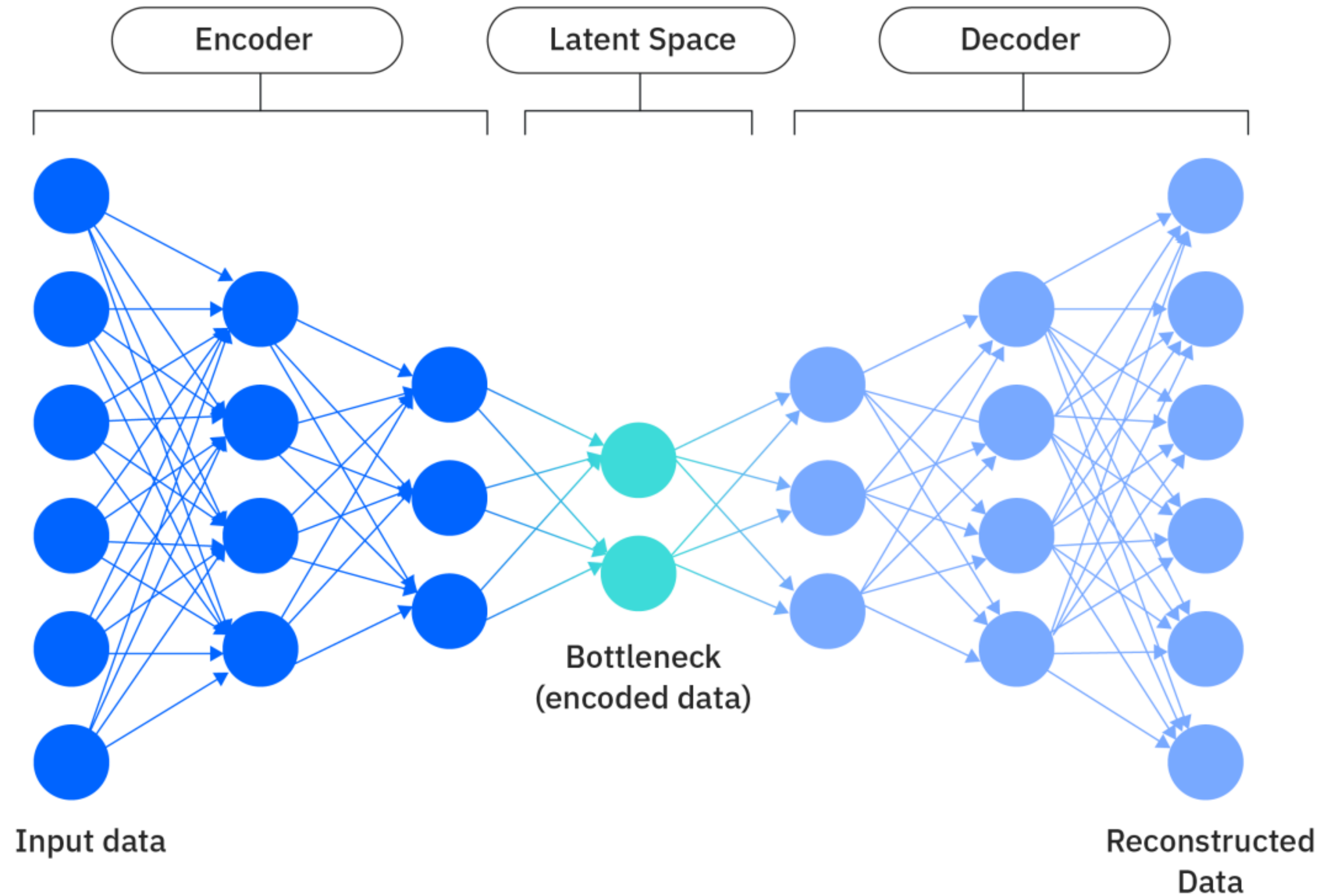


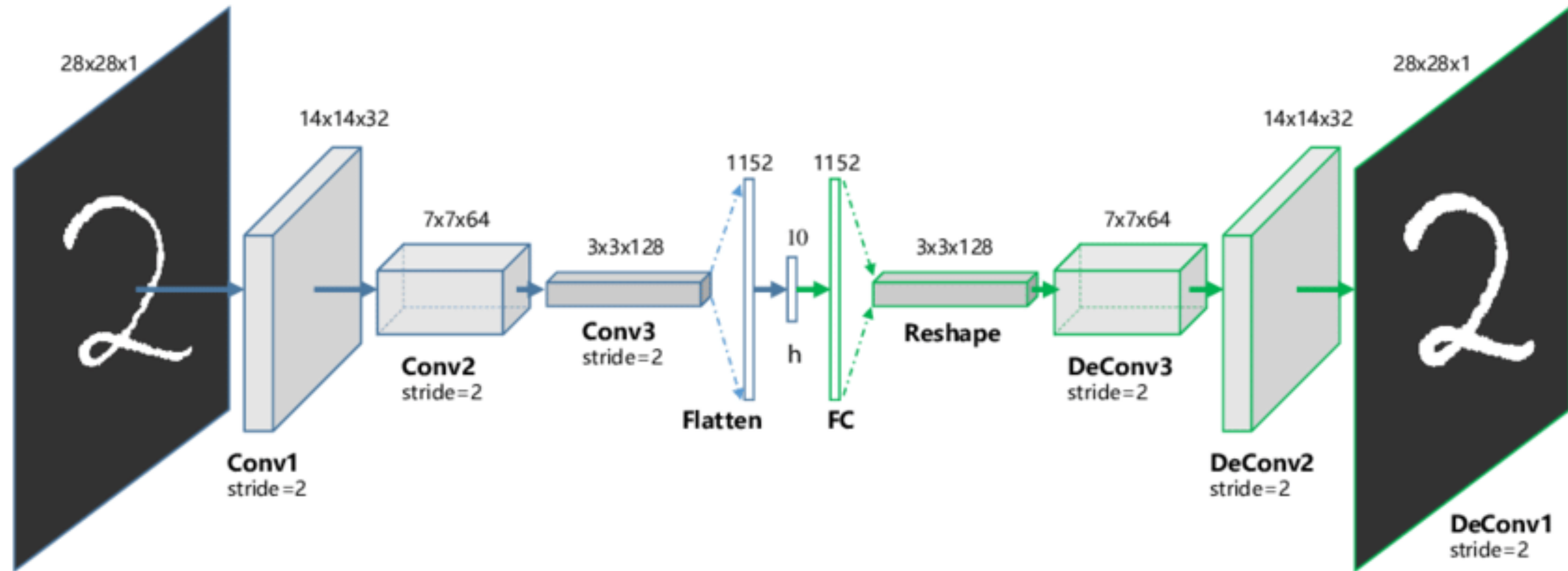
Audio Autoencoder

Autoencoder Architecture



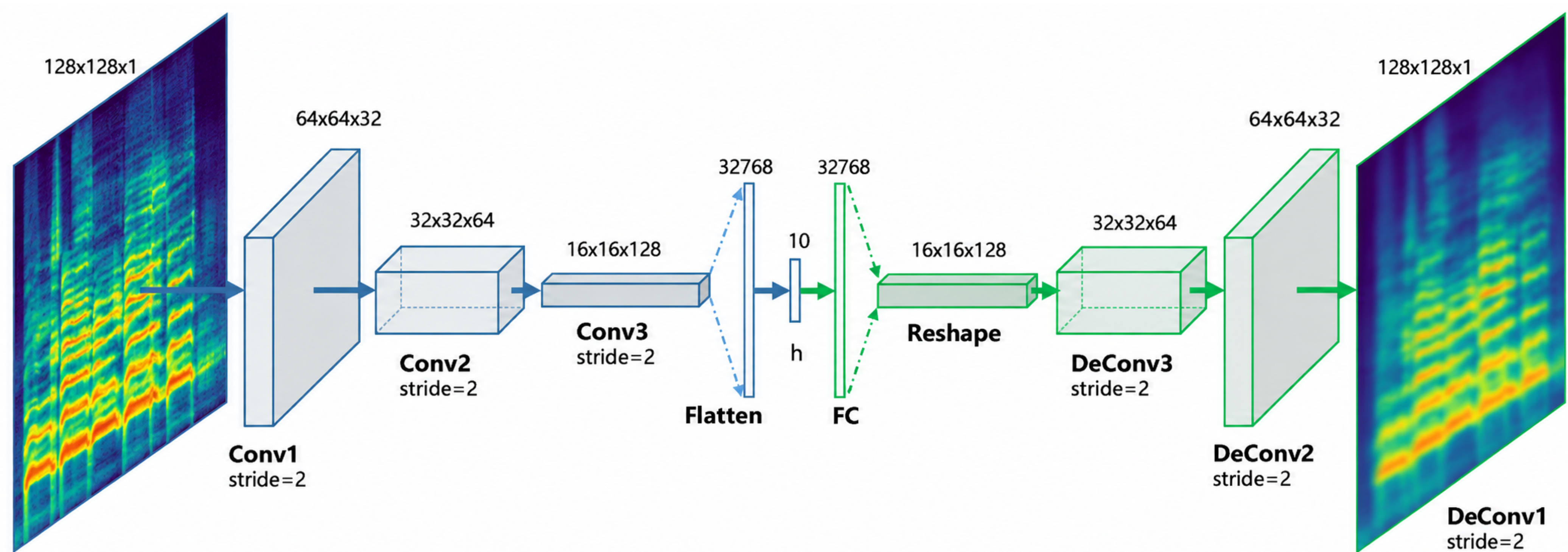
From: <https://www.ibm.com/think/topics/latent-space>

Convolutional Autoencoder



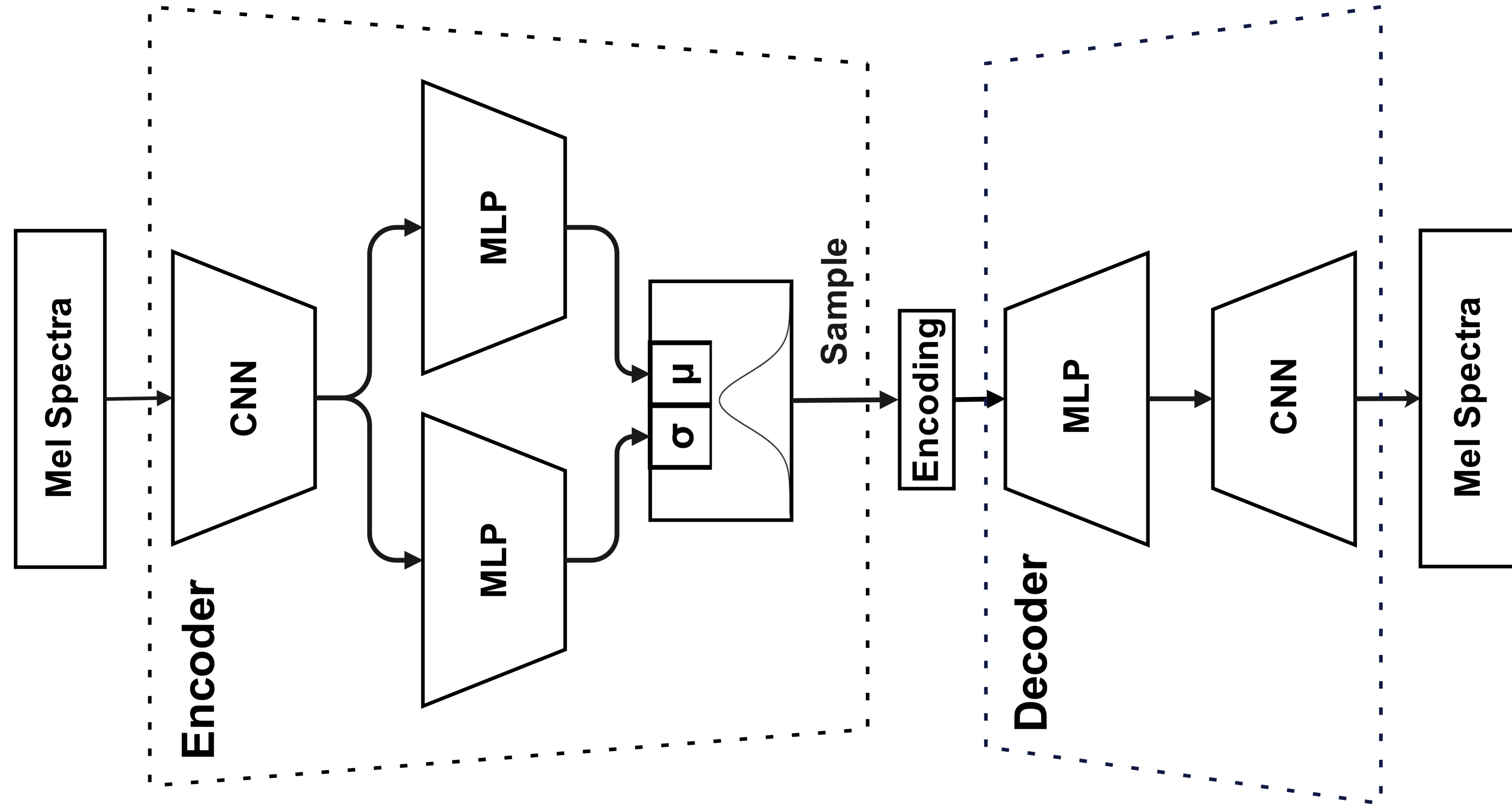
Convolution and Deconvolution Layers in an Autoencoder. ©Vaibhav Kumar

Convolutional Autoencoder

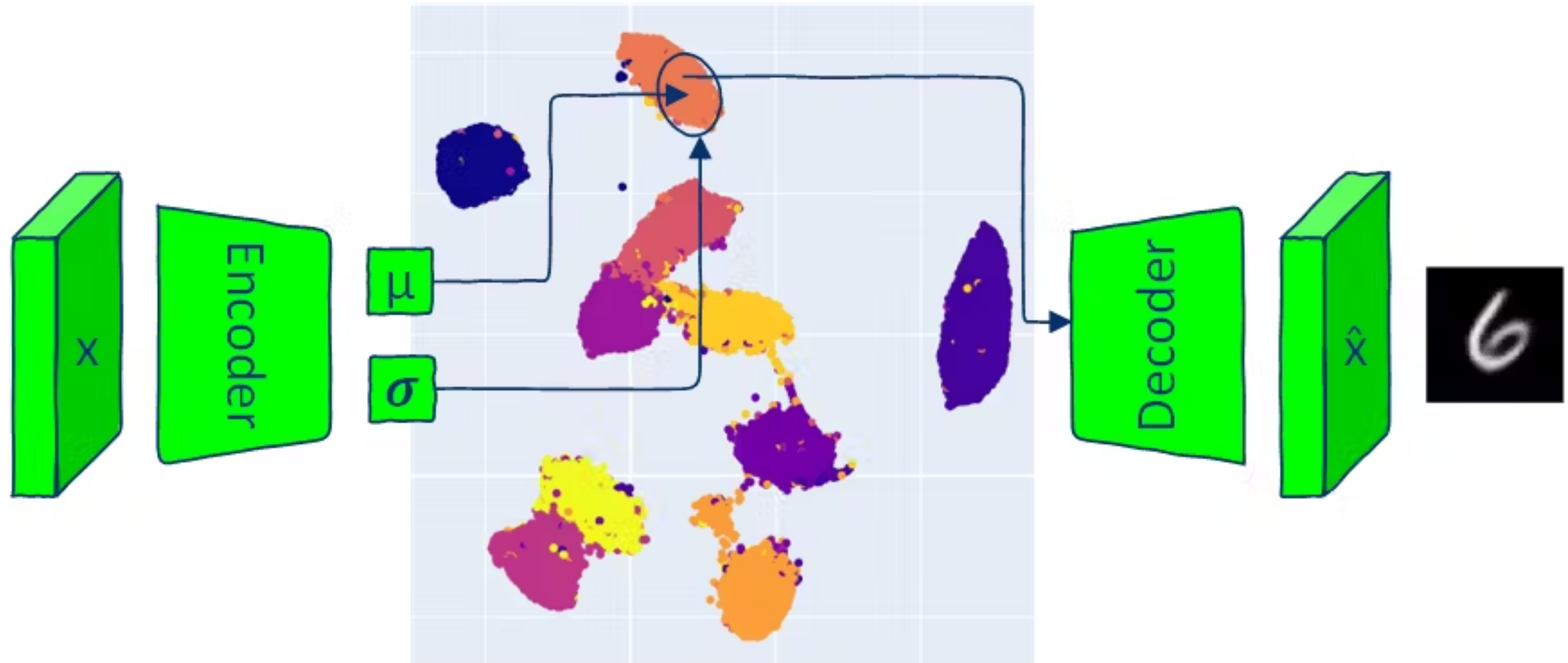


Convolution and Deconvolution Layers in an Autoencoder operating on Mel Spectra

Audio Autoencoder

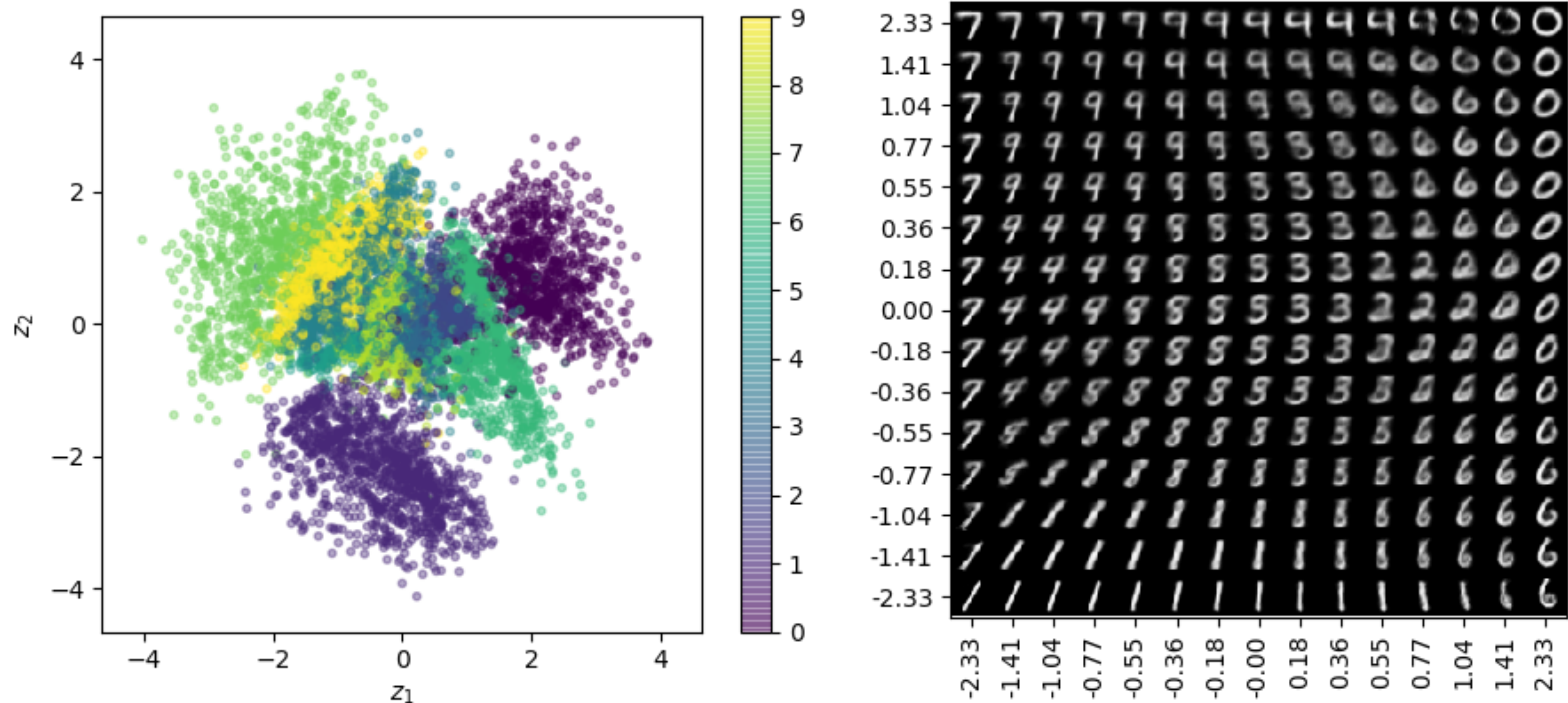


Variational Autoencoder



From <https://hackernoon.com/how-to-sample-from-latent-space-with-variational-autoencoder>

Latent Space

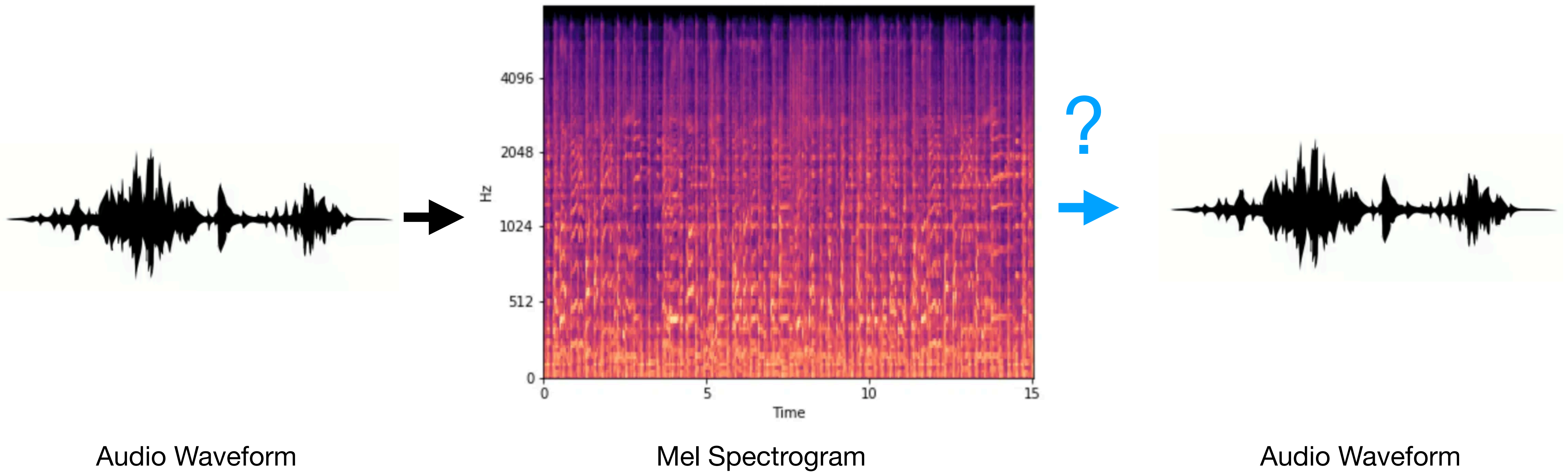


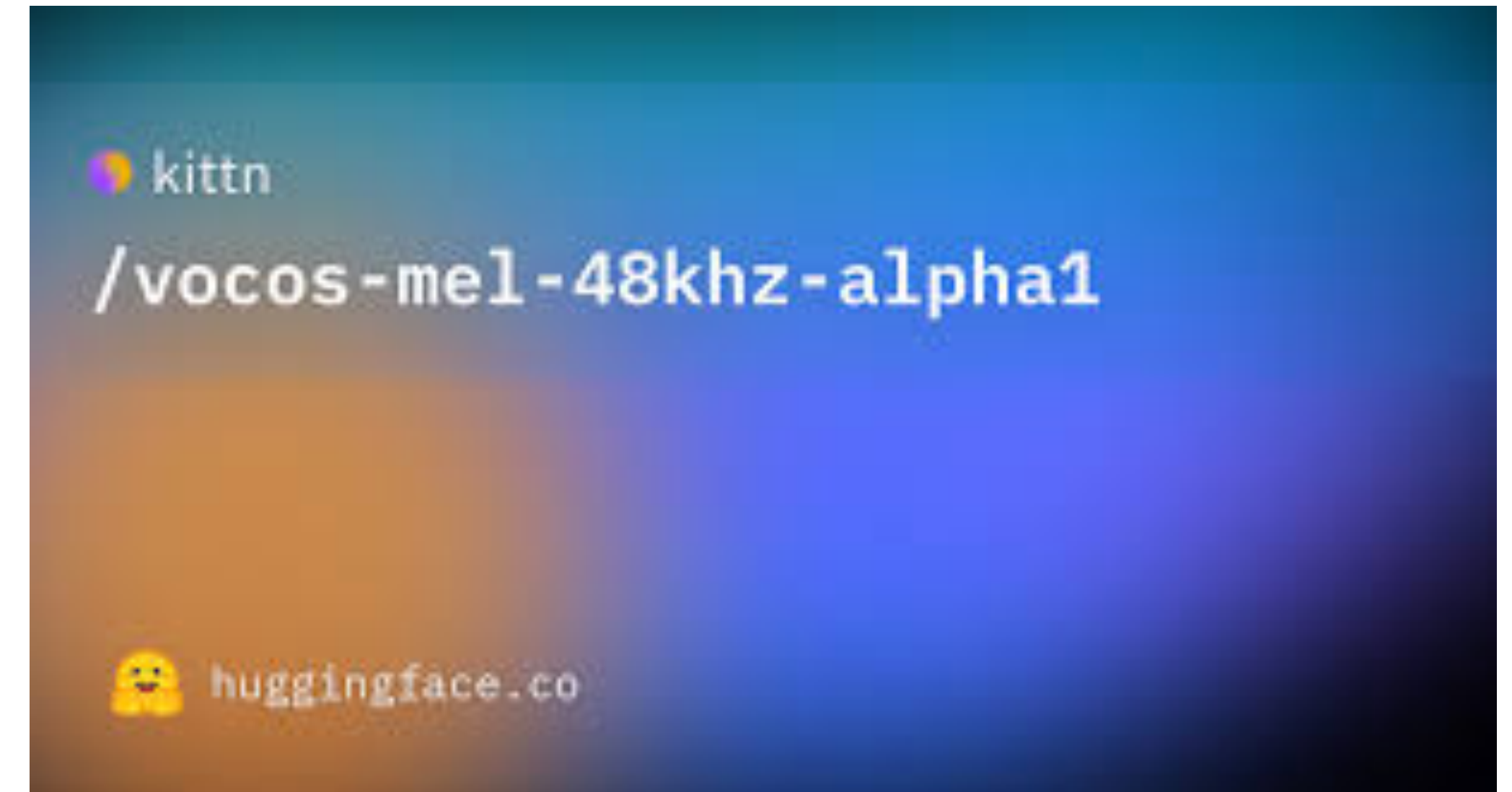
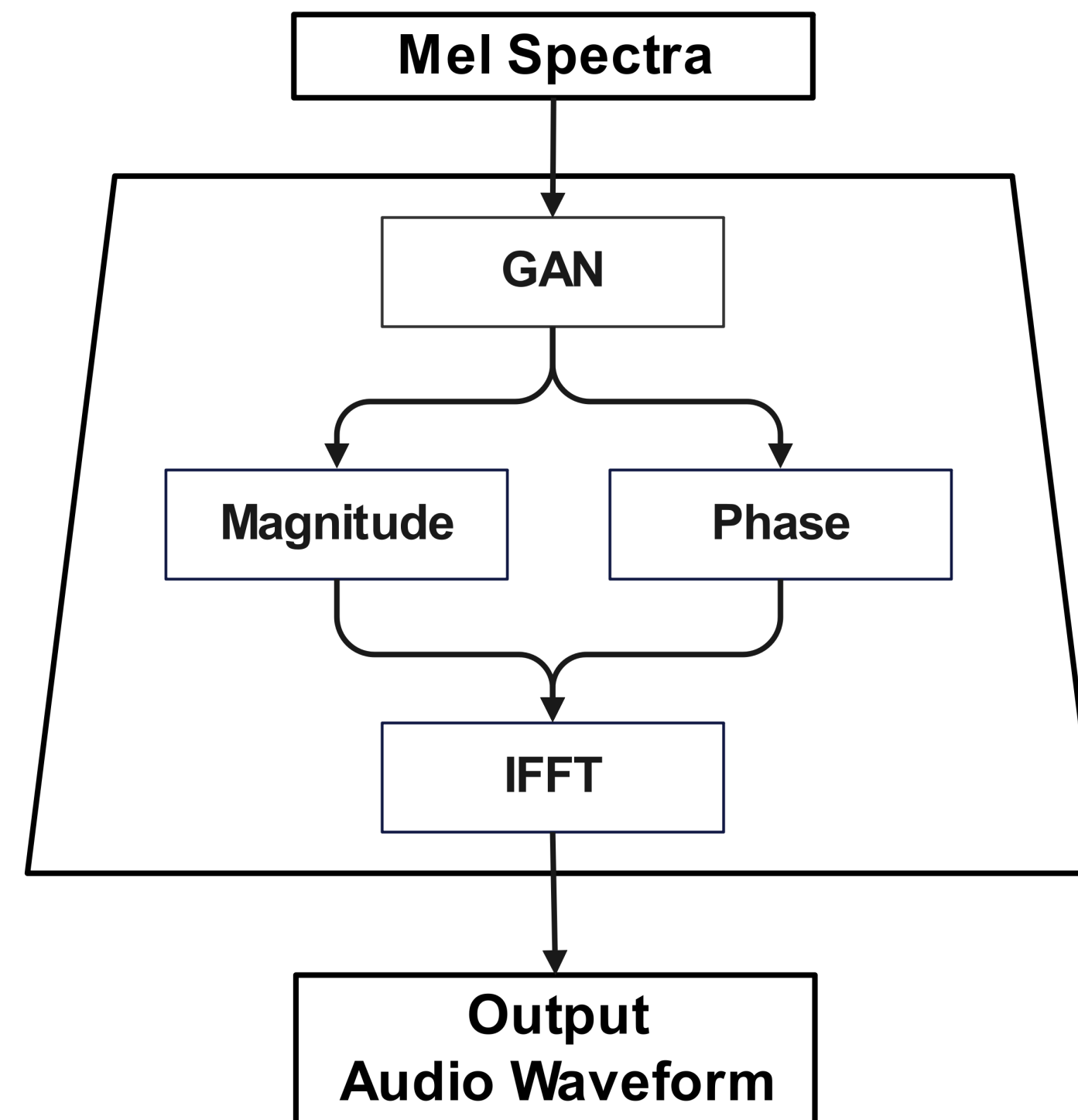
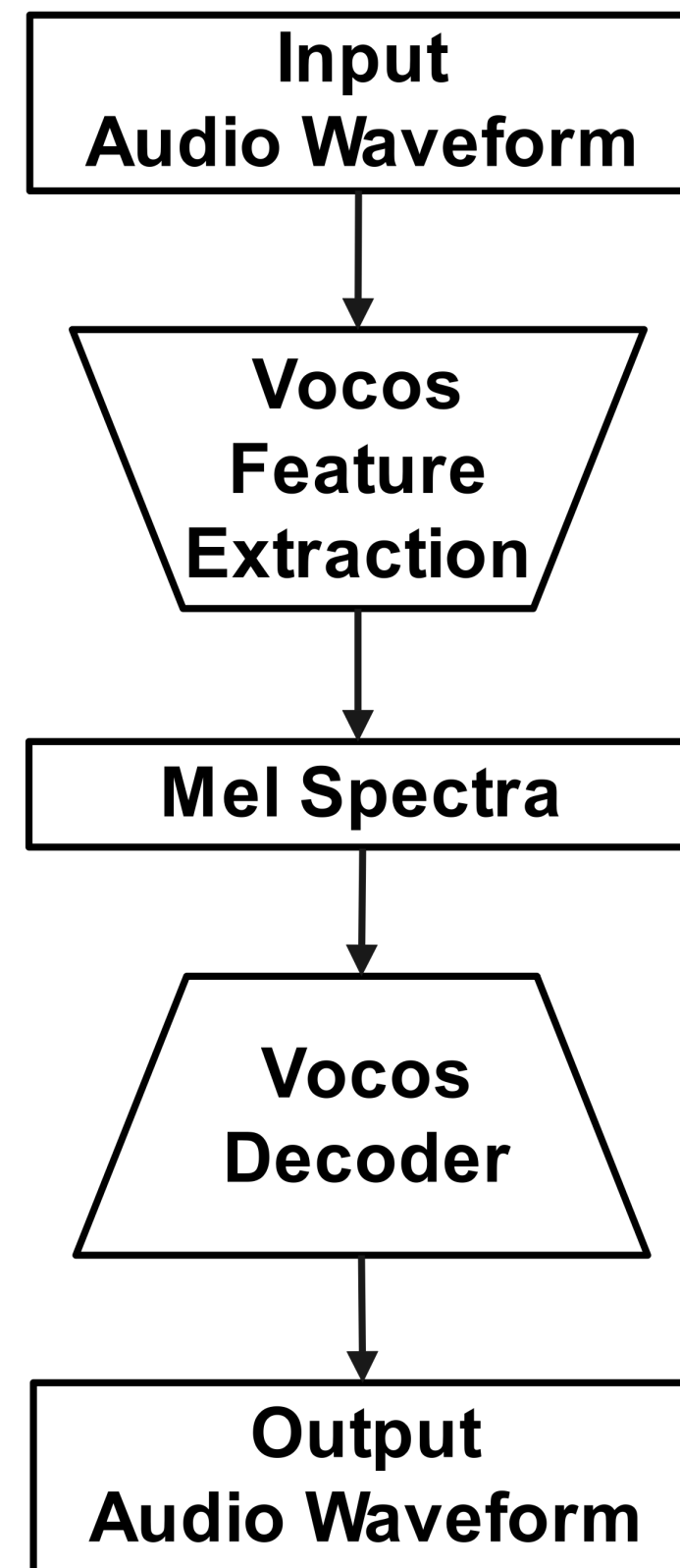
From: <https://tiao.io/post/tutorial-on-variational-autoencoders-with-a-concise-keras-implementation/>

Latent Space

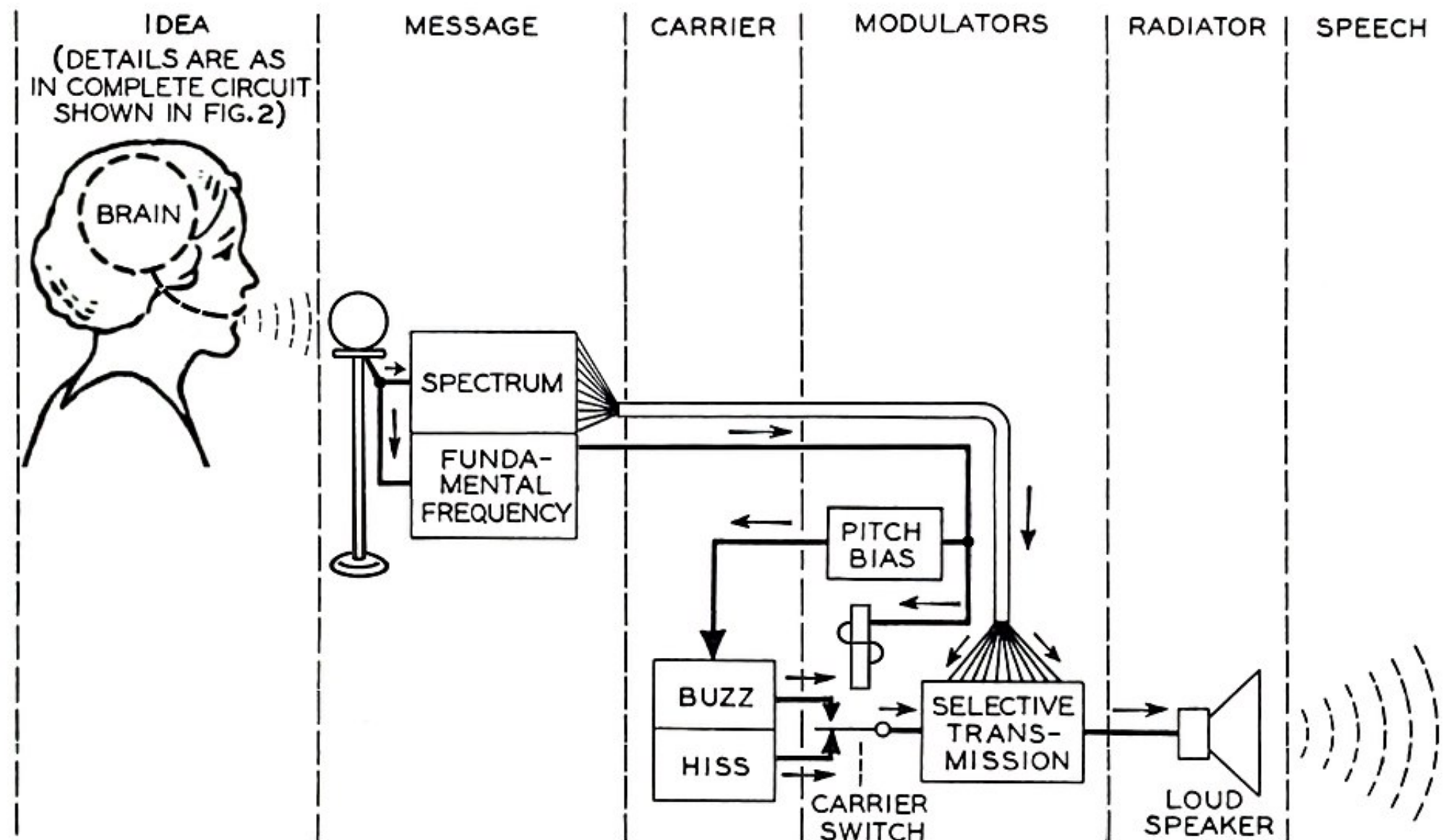


Bilinear Interpolation of Encodings of Images of Faces. ©Mathijs Pieters and Marco A. Wiering, 2018

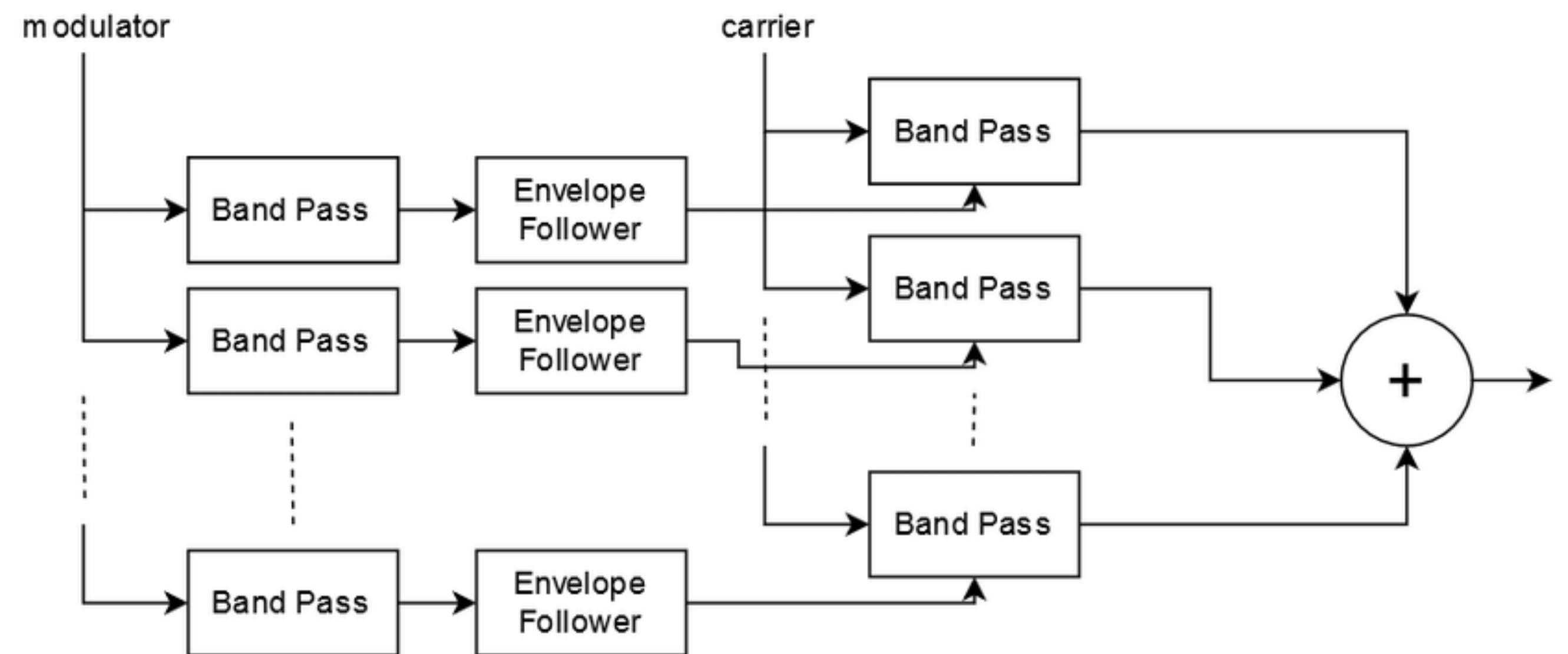




Audio Vocoder



Dudley's Vocoder, based on: [Dudley 1940](#)



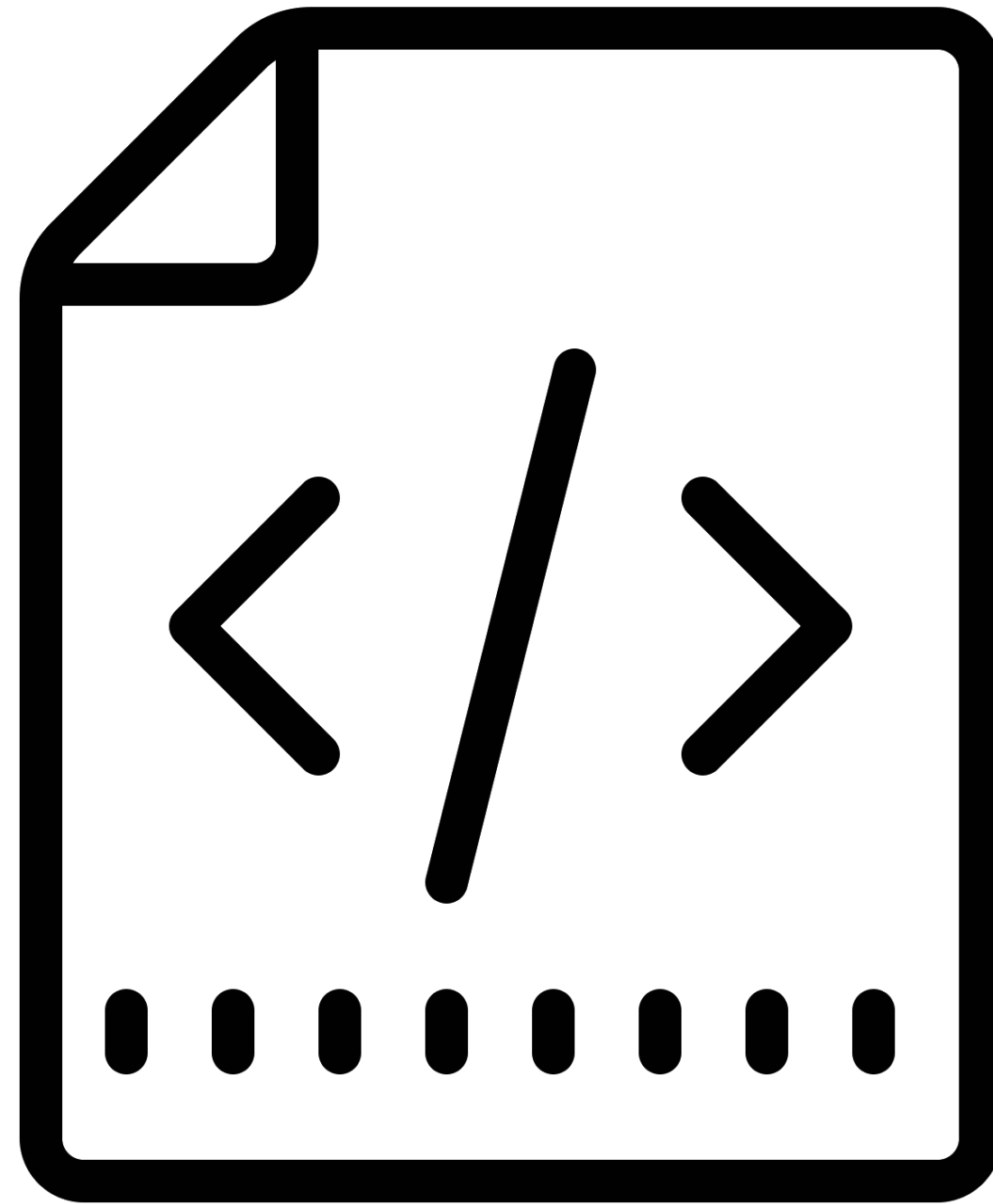
Schematic of a standard vocoder (Wikipedia)

Kraftwerk - Wir sind die Roboter (1978)

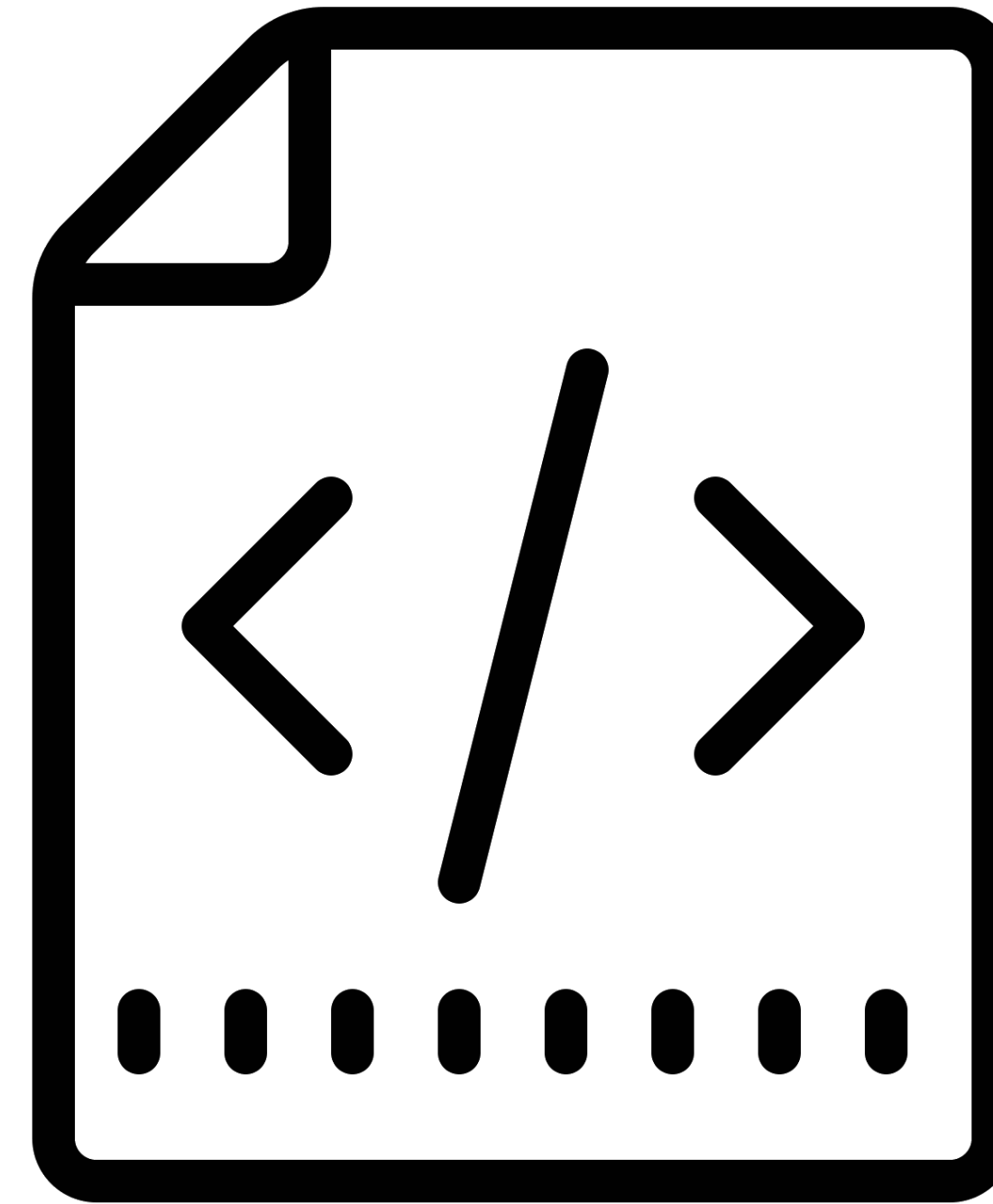


<https://www.youtube.com/watch?v=JgtFRj38fEw>

Audio Autoencoder - Demo - Training and Inference

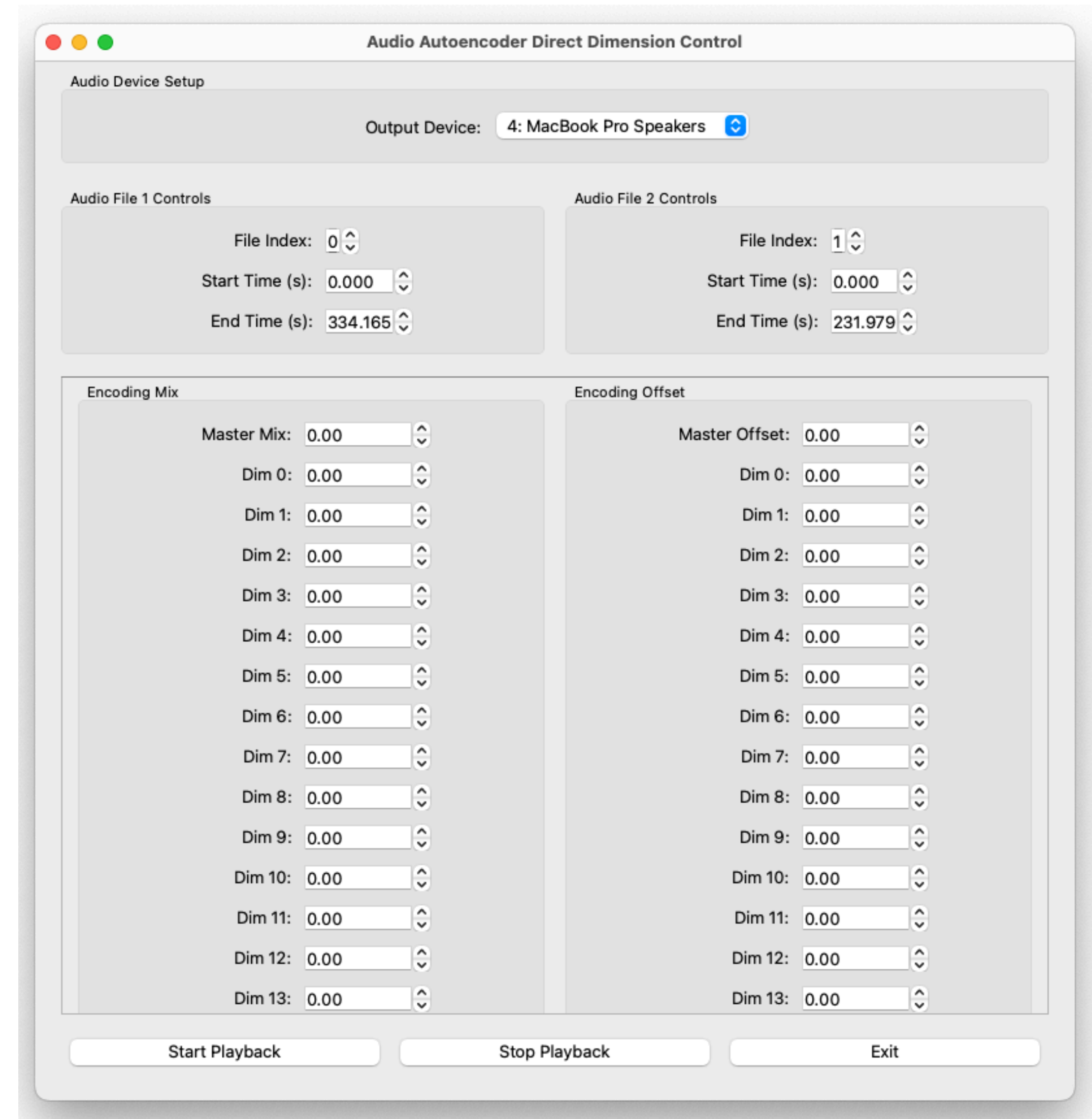
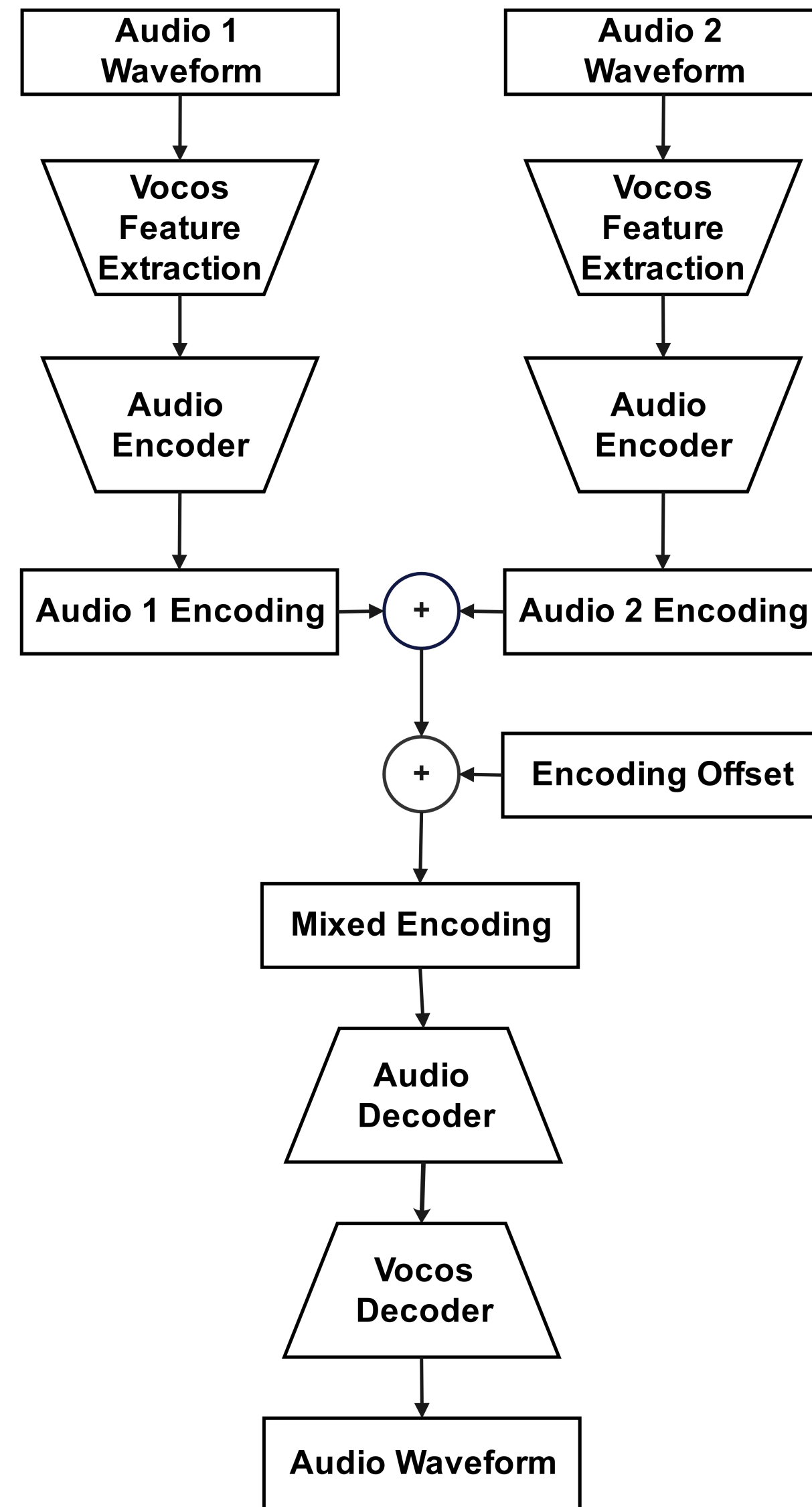


AI_Sound / Autoencoder / Training
audio_vae_vocos_cnn.py



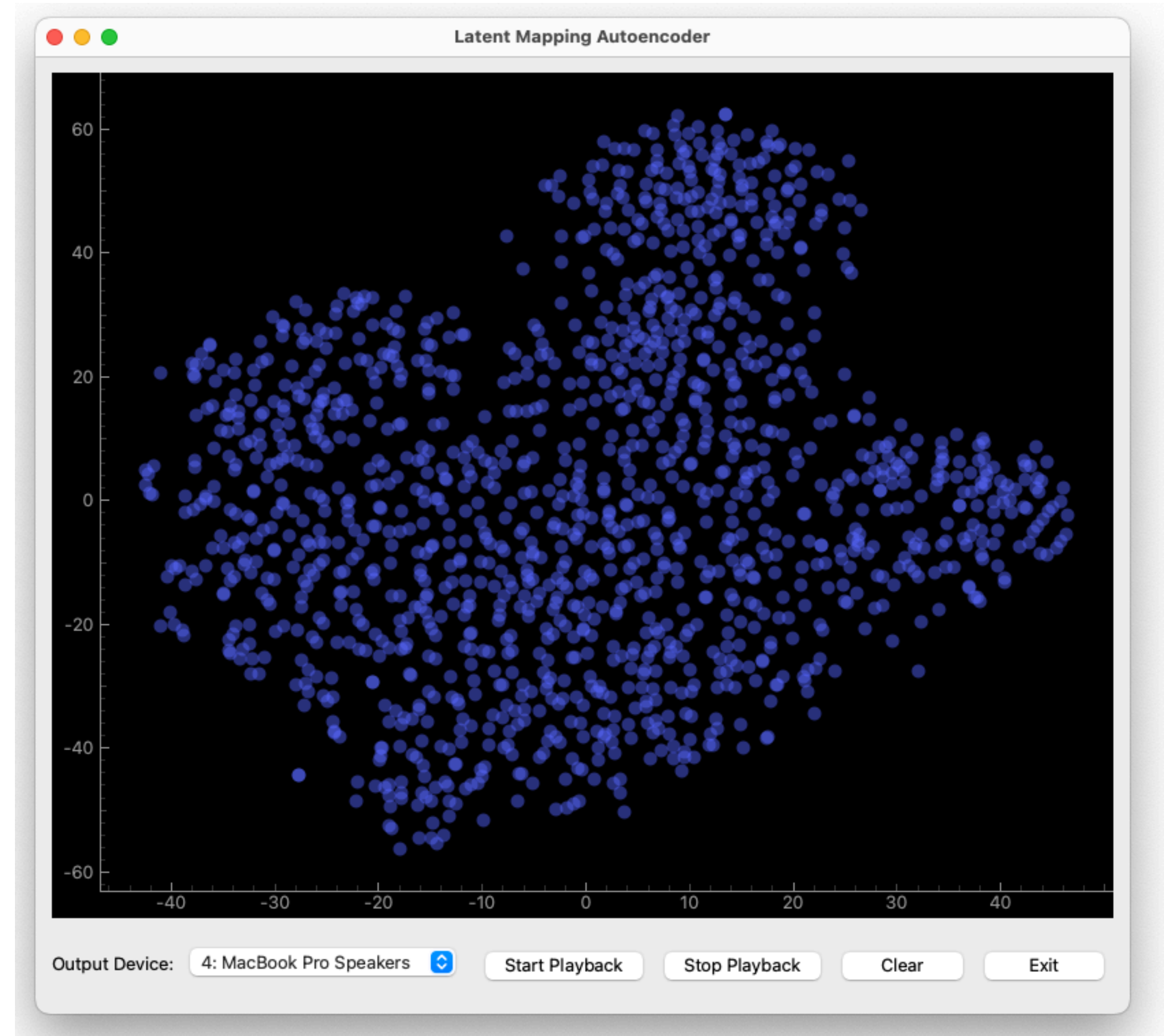
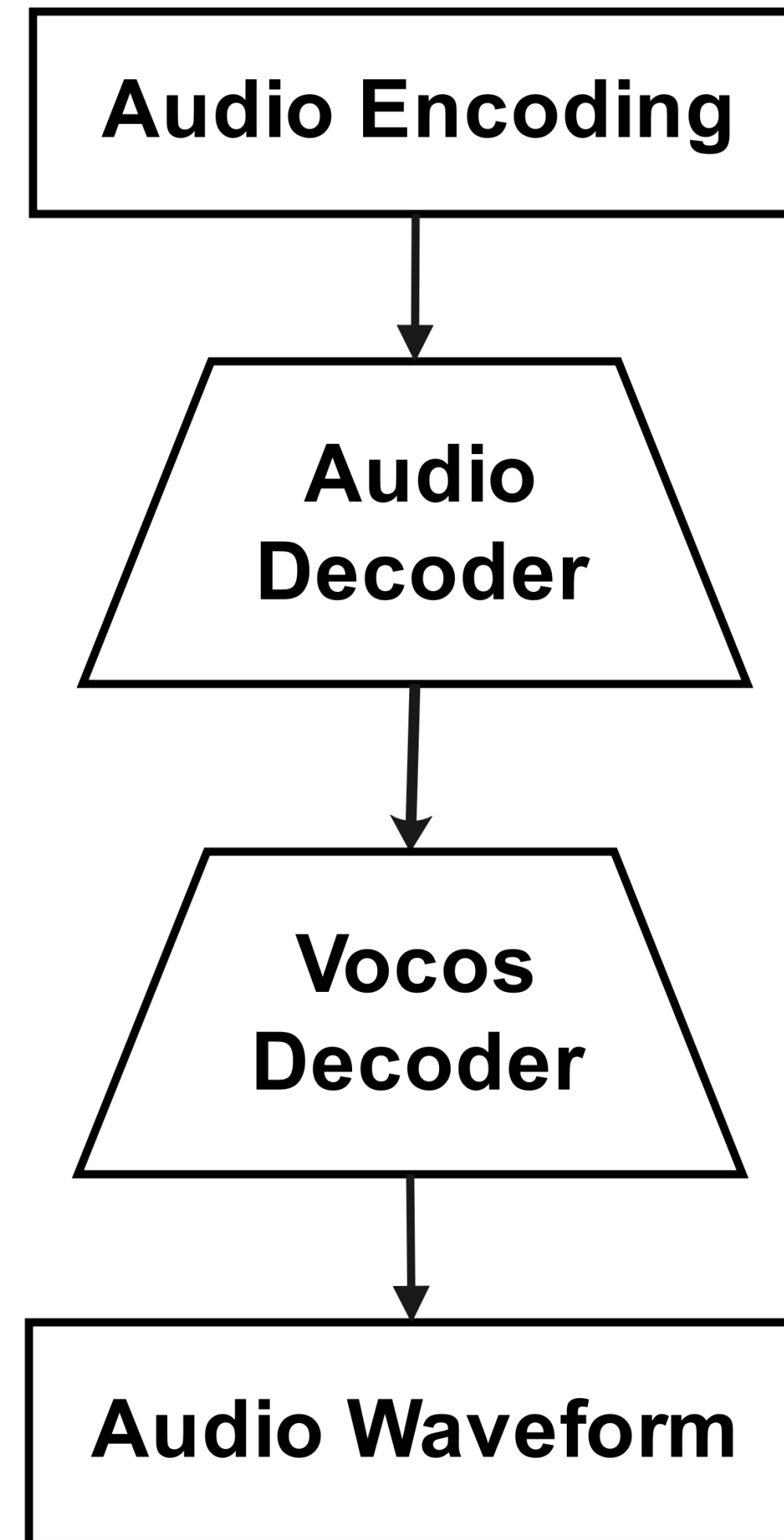
AI_Sound / Autoencoder / Inference / HelloWorld
vae_vocos_cnn_hello_world.py

Audio Autoencoder - Demo - Latent Mixing



AI_Sound / Autoencoder / Inference / LatentManipulation /
vae_vocos_cnn_interactive.py

Audio Autoencoder - Demo - Latent Navigation



AI_Sound / Autoencoder / Inference / LatentNavigation /
vae_vocos_cnn_interactive.py

Audio Autoencoder - Demo - Realtime Training

Live Audio VAE: train and listen

Data and training

Losses

Live inference

VAE mel MSE (Vocos mels)

Weight (0 = off): 0.5000

Running value: 0.47024 (weighted 0.23512)

Perceptual multi-res mel-STFT (auraloss)

Weight (0 = off): 0.5000

Mel bins: 128

Perceptual weighting: ☒

Running value: 1.27145 (weighted 0.63572)

Multi-resolution log-mel spectrogram

Weight (0 = off): 0.0000

Resolutions: 3

Smallest FFT size: 512

Mel bands: 80

Distance: L1

Running value: off

Multi-resolution log linear spectrogram

Weight (0 = off): 0.0000

Resolutions: 3

Smallest FFT size: 512

Pooled bands (0 = all bins): 0

step 313 (4.3 it/s) beta 0.1000 total 4.4755 kld 1.2130

vae_mel 0.4702 mr_stft 1.2714

audio queue 16/24

AI_Sound / Autoencoder / Training /
audio_vae_vocos_cnn_live_train.py

- Larger Audio Window
- Preserve Temporality in Latent Representation
- Deeper Architecture with Residual Connections
- Adversarial Training
- Vocos finetuning